

AI Companion: Technical Design

ProjectExtract, May 2026

Class layout

Class / asset	Role
ACompanionCharacter	Pawn. Owns weapon actor, health, traversal, anim BP.
ACompanionAIController	Possesses the pawn, runs the BT, binds external events to BB keys.
BT_Companion / BB_Companion	Behaviour tree + blackboard.
UHealthComponent	HP, shield, DBNO, revive interaction.
UTraversalComponent	Vault / climb / mantle. Shared with the player.
UBarkComponent	Event-driven bark playback, mode-filtered.
UCompanionTuningDataAsset	Designer-tunable formation, mirror, warp values.
UCompanionCharacterDataAsset	Per-mission identity (mesh, voice, name, portrait, per-character overrides).
UCompanionClassDataAsset	Optional stretch — class tuning preset.

Behaviour tree shape

Top-level Selector, left-to-right priority:

Priority	Branch	Decorator
1	Revive Player	PlayerDowned
2	Mirror Player Traversal	PlayerTraversalActive
3	Execute Player Command	CommandedAction != None
4	VIP Escort	EscortingVIP
5	Combat	HasEnemyTarget
6	Follow / Formation	(default, always succeeds)

Branch interrupts are decorator aborts (Self / Lower Priority / Both as appropriate). The aborts are what give the companion its responsiveness.

Condition flip	Effect
PlayerDowned → true	Combat / follow / command all abort. Revive takes over.

Condition flip	Effect
CommandedAction set	Combat / follow abort. Command takes over.
HasEnemyTarget → false	Combat aborts. Follow resumes.
EscortingVIP → true	Combat / follow continue, but inside the escort branch's distance constraint.
IsDowned → true (own)	All non-downed branches abort.

Blackboard schema

Key	Type	Set by	Read by
PlayerActor	Actor	AIController on possess	Revive, Follow, Command
PlayerDowned	Bool	Player HealthComponent event	Revive decorator
PlayerTraversalActive	Bool	Player TraversalComponent event	Mirror decorator
PlayerTraversalObsacle	Vector	Player TraversalComponent event	Mirror task
PlayerTraversalLanding	Vector	Player TraversalComponent event	Mirror task
CommandedAction	Enum	Player Controller (ping / wheel)	Command decorator
CommandTargetActor	Actor	Player Controller	Command tasks
CommandTargetLocation	Vector	Player Controller	Command tasks
CompanionMode	Enum	Player input / auto-suggest acceptance	All mode-aware tasks
SuggestedMode	Enum	Auto-suggest service	HUD only
HasEnemyTarget	Bool	Combat service	Combat decorator
EnemyTarget	Actor	Combat service	Combat tasks
CurrentCoverSlot	Actor	MoveToCover task	Combat sub-tree, cover-quality service
CoverStillGood	Bool	Cover-quality service	Combat sub-tree

Key	Type	Set by	Read by
EscortingVIP	Bool	Mission setup	VIP decorator
VIPActor	Actor	Mission setup	VIP tasks
IsDowned	Bool	Own HealthComponent	Top-level decorator
TargetMarked	Actor	Player ping in Stealth	Stealth fire gate

Mode system

CompanionMode is a single enum on the blackboard, read at the point of decision rather than branched on at the top of the BT. Adding a new mode-aware behaviour means updating one task, not restructuring the tree.

Per-task modulation:

Behaviour	Combat	Stealth	Explore
Move speed	Sprint between cover, walk in cover	Walk only	Walk near player
Fire policy	Any enemy in LoS	Hold unless TargetMarked	Any enemy in LoS (auto-flips to Combat)
Audio policy	Full bark bank	Whisper subset only	Conversational subset
Cover policy	Default; opportunistic reposition	Sticky unless undetected	Loose; near player
Enemy perception	Standard	Visual ignored, sound only	Standard

Auto-suggest

Lives on the player controller. Watches gameplay events, writes SuggestedMode (not CompanionMode). HUD reads SuggestedMode and shows a toast. Player acceptance writes the mode change to CompanionMode. Suggestion never bypasses the player.

Trigger	Suggested mode
Unsilenced fire while enemies aware	Combat
Crouch + suppressor + no enemies alerted	Stealth
No enemies present for sustained period	Explore

Command surface

Three input methods routed to the same blackboard keys.

- **Ping** — camera trace from player controller, resolves to a `CommandedAction` based on what the trace hit and the current mode.
- **Hotkeys** — mode set (1/2/3) and regroup (G or similar).
- **Wheel** — held key, radial menu for commands needing a menu pick.

Ping resolution

Trace hit	Combat / Explore	Stealth
Enemy actor	Focus Fire	Mark Target (paired silent fire on player's shot)
Lootable actor	Mark Loot	Mark Loot
Ground	Move to Point	Move to Point
Door / interactable	Breach	Breach
Nothing in range	Hold Position at cursor	Hold Position at cursor

Persistence

- **Persistent state:** `SetMode`, `HoldPosition`. Held until cleared by a new command.
- **One-shot:** everything else. Complete, clear `CommandedAction`, fall through to lower-priority branches.

The command branch is a Selector of children, each gated on a specific `CommandedAction` value.

Cover system

Authored. `CoverSlot` actors carry peek metadata. `CoverRegion` actors group slots so the AI rotates between sub-slots within a region rather than committing to one spot for a whole engagement.

Cover selection EQS

`EQS_Companion_Cover` query, generated against slots in radius around the player. Weighted tests:

Test	Weight	Effect
LoS to enemy	High	Slot useless without it
Distance to player	Medium	Penalises too-far slots
Peek direction match	High	Authored peek faces the threat
Not in teammate fire arc	High	Avoids being shot in the back
Slot height vs threat	Medium	Reject cover too short to fire over
Distance from current slot	Low	Mild stickiness

Highest score wins.

Cover-still-good check

Service running ~1 Hz re-evaluates the current slot against the same tests. Score drop below threshold flips `CoverStillGood` to false; Combat sub-tree picks a new slot. This is the engine behind the opportunistic-reposition behaviour described in the gameplay overview.

Safety nets

- **Watchdog**: timeout on path-to-slot. Picks a different slot or warps on expiry.
- **Reload gate**: no fresh peek cycle while reloading.

VIP escort

Branch slotted between command and combat priority. Active only when `EscortingVIP` is set (mission setup writes it at start).

- **Default**: companion holds within configurable distance of `VIPActor`. Cover EQS adds a "distance to VIP" weighted test on top of the standard cover scoring.
- **VIP follow leader** is the companion. A "Stay With Me" command from the player flips the VIP's leader to the player; reverse flips back. Single boolean swap, no new behaviour.
- **Companion-downed** → **VIP cower**: `HealthComponent` fires `CompanionDowned` event to `GameState`. VIP switches to cower-in-place idle. State clears on companion revive.
- **Mission failure** held on the `GameMode`. Listens for both player-DBNO and companion-DBNO events. Concurrent = checkpoint reload.

Vulnerability

`HealthComponent` state machine:

State	Entered when	Behaviour
Alive	Spawn / Revived	Standard combat
ShieldBroken	Shield at 0	HP takes damage. Shield-regen recovery timer.
Downed	HP at 0	Combat / command branches abort via <code>IsDowned</code> . Bleed timer + revive interaction exposed.
Dead	Bled out without revive	Only the player normally reaches this. Mission failure path.

Damage routes through `TakeDamage`: shield first, then HP. Revive task is shared between player and companion via a `RevivingActor` key the target's `HealthComponent` watches.

Threat priority

Per-enemy, per-tick perception service. Scores candidate targets:

Input	Weight	Effect
Distance to target	High	Closer = higher
Recency of damage from target	High	Target who just shot us = higher
Target exposed vs in cover	Medium	Exposed = higher
Companion in Stealth + no alert	Override	Zeros companion's score
Target killed teammate recently	Medium	Bumps

Enemies engage the highest scorer. The Stealth override implements the "invisible to vision" rule from the gameplay overview — the companion can still be heard, but can't be scored on visual perception alone.

Class / character system

DataSet-driven. No code change per character.

`CompanionCharacterDataSet` **(per-mission)**

- Skeletal mesh
- Animation blueprint
- Voice bank (bark audio)
- Display name + portrait
- Per-character overrides (accuracy ramp, reaction time, etc.)

`CompanionClassDataSet` **(stretch)**

- Default loadout (weapon class, ammunition type, abilities)
- HP / shield baselines
- Accuracy ramps
- Cover preferences (a stealth class biases against noisy positions like windows)
- Default mode bias

The class is an additional layer; player commands still override class defaults at runtime.

`CompanionTuningDataSet` already exists and holds formation, mirror, and warp values. Character + class assets are the next layer up.

Adding a new companion = one new character asset, optionally a new class asset. Nothing else changes.

Communication

Bark pipeline

Event source → `UBarkComponent` → bucket lookup → mode-filtered set → random pick → play.

Event source	Bucket
Perception OnEnemySpotted	Threat alert
Weapon OnReloadStart	Self-state
HealthComponent OnDowned	Self-state
AIController OnCommandReceived	Command response
Mission OnObjectiveUpdated	Mission-state

If the mode-filtered set is empty (no whisper variant of this line exists for stealth, say), no audio plays. HUD layer can still surface the event silently.

Enemy-spotted marker

Separate from barks. Perception broadcasts a spotted-target event through the AIController; HUD draws a world-space marker pinned to the enemy. Fades after a few seconds, refreshes on new sighting. **Load-bearing scout function** — barks are flavour, this carries the actual position information.

HUD widgets

Delegate-bound, no per-frame polling.

Widget	Bound to
HP / Shield bar	HealthComponent OnHealthChanged
Mode indicator	CompanionMode (replicated to player)
Command-state indicator	CommandedAction
Low-ammo warning	Weapon OnLowAmmo
Mode-suggestion toast	SuggestedMode (replicated)

Stretch goal mapping

Stretch	Architecture cost
Class system	Add CompanionClassDataAsset, route through existing tuning hooks. Data only.
Second companion slot	TeamRoster array on GameState. Per-companion AIController + BT instance. Arbitration logic for command dispatch and revive priority. High.
Walk-up silent knife kill	New BT task gated by TargetMarked && CompanionMode == Stealth && Distance < Threshold. Animation pair (companion + victim). Low-medium.

Stretch	Architecture cost
Persistent character recurrence	Save-game flag, per-character reunion bark line. Low.
Banter / downtime lines	New bucket in bark bank, timer service in Explore mode. Low.
Autonomous tactical combat	Tactical decision service in Combat sub-tree. New flank / push / retreat sub-trees. Cover system already supplies the destinations. High.

Build state

Prototype runs end-to-end: follow, combat, cover (authored slots + peek matrix + walk-approach), traversal (mirror + warp safety nets + falling locomotion), revive. Recent work since the last status doc: reload gates, LoS-gated targeting, watchdog timers, mesh-attached weapon, HP widget anchor.

Outstanding implementation, in order

#	System	Notes
1	Cover progression	Inter-slot movement triggered by player movement
2	Mode system	CompanionMode BB key + auto-suggest service on player controller
3	Command surface	Ping resolution, wheel, BT command branch
4	Companion DBNO	Mirror of player DBNO in HealthComponent
5	Player-side revive interaction	Walk-up + hold-key flow
6	VIP mission setup	Escort branch, VIP cower state, fail-on-team-wipe check
7	HUD layer	Markers, indicators, command-state widget
8	Bark system	UBarkComponent + first character's VO
9	Tutorial integration	First mission's in-fiction lesson triggers

Each is independent against the existing prototype.